Concavity, Curve Sketching, and Optimization

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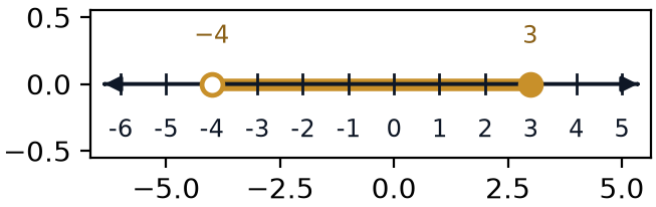
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Increasing/decreasing and the first derivative test

For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.

2 [Foundational] • Target: Concavity and the second derivative test

For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.



3 [Foundational] • Target: Concavity and the second derivative test

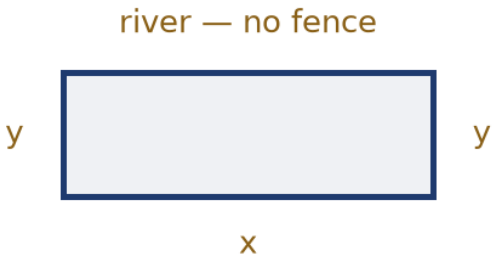
Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 3x$.

4 [Foundational] • Target: Curve sketching

For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.

5 [Foundational] • Target: Optimization problems

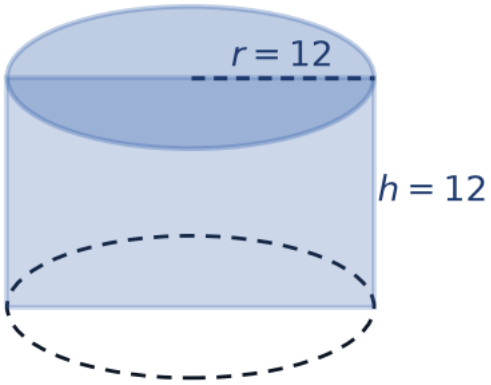
A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.



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6 [Foundational] • Target: Optimization problems

An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.



7 [Foundational] • Target: Optimization: minimizing cost or distance

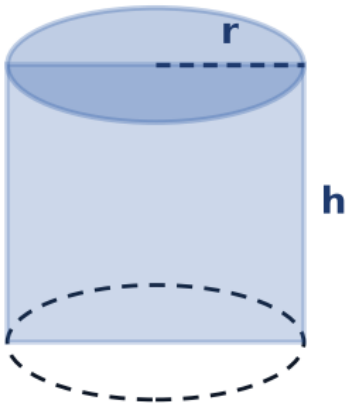
Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Optimization: minimizing cost or distance

A closed cylindrical can must have volume 16π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes surface area.



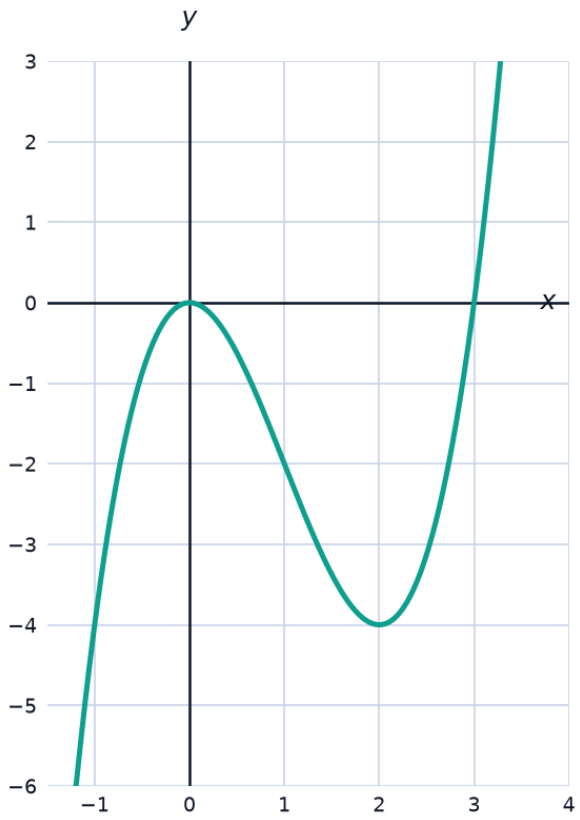
9 [Intermediate] • Target: Points of inflection and graph behavior

Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

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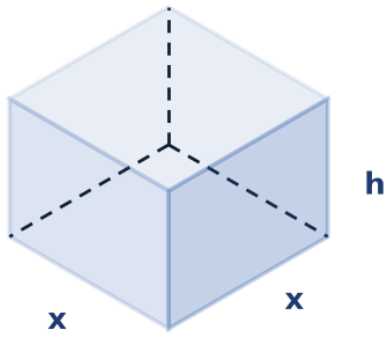
10 [Intermediate] • Target: Visualizing extrema and concavity together

The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.



11 [Intermediate] • Target: A capstone optimization problem

A rectangular box with a square base and no top must have volume 500 cm^3. Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.



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12 [Intermediate] • Target: Second derivative test failures

For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.

13 [Intermediate] • Target: A capstone curve-sketching problem

For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.

14 [Intermediate] • Target: Applied Optimization: Cost & Distance

Find the nonzero inflection point of $f(x) = x^4 - 4x^3$.

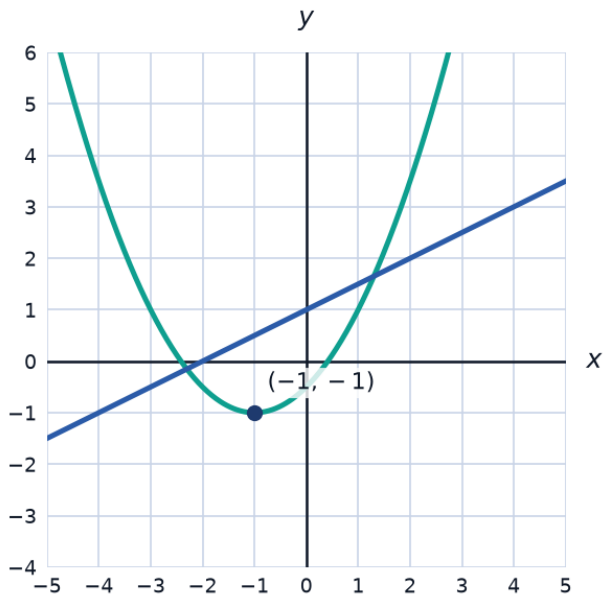
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Increasing/decreasing and the first derivative test • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find the nonzero inflection point of $f(x) = x^4 - 10x^3$.

16 [Challenging] • Target: Concavity and the second derivative test • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find the nonzero inflection point of $f(x) = x^4 + 2x^3$.



17 [Challenging] • Target: Curve sketching • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the nonzero inflection point of $f(x) = x^4 - 16x^3$.

18 [Challenging] • Target: Optimization problems • Cross-Lesson: 8.01 Area Between Curves

Find the nonzero inflection point of $f(x) = x^4 + 8x^3$.

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19 [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

A farmer has 100 m of fencing to enclose a rectangular field with $x + 2y = 100$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

20 [Challenging] • Target: Points of inflection and graph behavior • Cross-Lesson: 1.01 Limits Graphically and Numerically

A farmer has 72 m of fencing to enclose a rectangular field with $x + 2y = 72$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Increasing/decreasing and the first derivative test

Classify the critical point $x = -1$ of $f(x) = x^3 + 6x^2 + 9x$, using the Second Derivative Test.

a. a local maximum

b. cannot be determined without more information

c. a local minimum

d. an inflection point

22 [Foundational] • Target: Concavity and the second derivative test

Classify the critical point $x = 1$ of $f(x) = x^3 + 3x^2 - 9x$, using the Second Derivative Test.

a. cannot be determined without more information

b. a local minimum

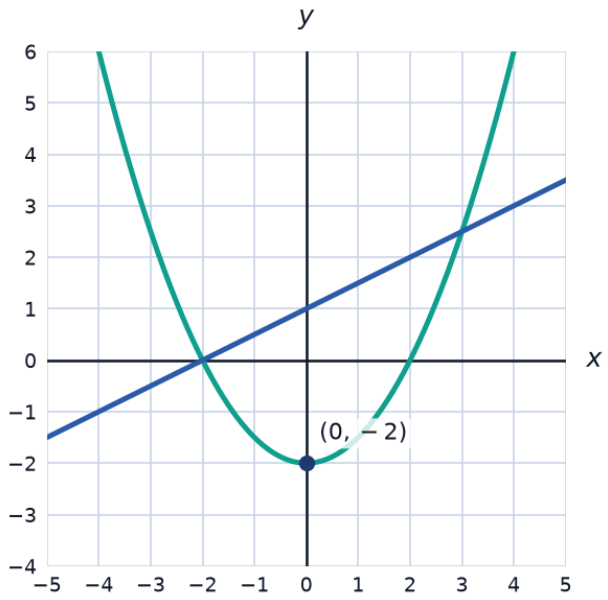
c. an inflection point

d. a local maximum

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23 [Foundational] • Target: Curve sketching

Classify the critical point $x = -6$ of $f(x) = x^3 + 12x^2 + 36x$, using the Second Derivative Test.



a. a local minimum

b. a local maximum

c. an inflection point

d. cannot be determined without more information

24 [Foundational] • Target: Optimization problems

Classify the critical point $x = 0$ of $f(x) = x^3 - 9x^2$, using the Second Derivative Test.

a. a local maximum

b. an inflection point

c. cannot be determined without more information

d. a local minimum

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25 [Foundational] • Target: Optimization: minimizing cost or distance

Classify the critical point $x = 6$ of $f(x) = x^3 - 3x^2 - 72x$, using the Second Derivative Test.

a. cannot be determined without more information

b. an inflection point

c. a local maximum

d. a local minimum

26 [Foundational] • Target: Points of inflection and graph behavior

Classify the critical point $x = 4$ of $f(x) = x^3 - 9x^2 + 24x$, using the Second Derivative Test.

a. a local maximum

b. an inflection point

c. a local minimum

d. cannot be determined without more information

27 [Foundational] • Target: Visualizing extrema and concavity together

Classify the critical point $x = -6$ of $f(x) = x^3 + 9x^2$, using the Second Derivative Test.

a. an inflection point

b. a local minimum

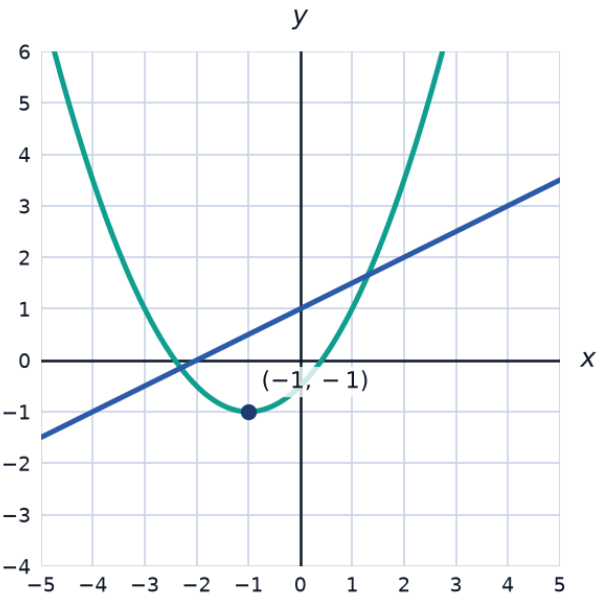
c. a local maximum

d. cannot be determined without more information

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28 [Foundational] • Target: A capstone optimization problem

Classify the critical point $x = -5$ of $f(x) = x^3 + 3x^2 - 45x$, using the Second Derivative Test.



a. an inflection point

b. a local maximum

c. cannot be determined without more information

d. a local minimum

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Second derivative test failures

Find the nonzero inflection point of $f(x) = x^4 + 14x^3$.

a. 14

b. -6

c. -14

d. -7

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30 [Intermediate] • Target: A capstone curve-sketching problem

Find the nonzero inflection point of $f(x) = x^4 - 2x^3$.

a. -2

b. 2

c. 4

d. 1

31 [Intermediate] • Target: Concavity & Points of Inflection

Find the nonzero inflection point of $f(x) = x^4 - 14x^3$.

a. 8

b. 14

c. 7

d. -14

32 [Intermediate] • Target: Second Derivative Test for Local Extrema

Find the nonzero inflection point of $f(x) = x^4 - 12x^3$.

a. 7

b. 6

c. -12

d. 12

33 [Intermediate] • Target: Applied Optimization: Geometric Problems

Find the nonzero inflection point of $f(x) = x^4 - 8x^3$.

a. 5

b. 8

c. -8

d. 4

34 [Intermediate] • Target: Applied Optimization: Cost & Distance

A farmer has 108 m of fencing to enclose a rectangular field with $x + 2y = 108$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

a. 28

b. 108

c. 27

d. 54

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35 [Intermediate] • Target: Increasing/decreasing and the first derivative test

An open-top box is made from a 36 in by 36 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(36 - 2x)^2$.

a. 6

b. 18

c. 9

d. 7

36 [Intermediate] • Target: Concavity and the second derivative test

An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(12 - 2x)^2$.

a. 4

b. 2

c. 3

d. 6

37 [Intermediate] • Target: Curve sketching

An open-top box is made from a 48 in by 48 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(48 - 2x)^2$.

a. 9

b. 8

c. 24

d. 12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Optimization problems • Cross-Lesson: 8.01 Area Between Curves

Find the x-coordinate of the point on the line $y = 4x + 5$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. $-\frac{5}{17}$

b. $-\frac{5}{4}$

c. $-\frac{20}{17}$

d. $\frac{20}{17}$

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39 [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Find the x-coordinate of the point on the line $y = x + 3$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. $-\frac{3}{2}$

b. -3

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

40 [Challenging] • Target: Concavity & Points of Inflection • Cross-Lesson: 1.01 Limits Graphically and Numerically

On which interval is the function $f(x) = x^3 - 6x^2 + 9x$ concave up?

a. $(2, \infty)$

b. $(-\infty, 2)$

c. $(1, 3)$

d. $(0, \infty)$

41 [Challenging] • Target: Concavity & Points of Inflection • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

What is a necessary condition for $(c, f(c))$ to be a point of inflection?

a. $f''(c) = 0$ or $f''(c)$ undefined, AND f'' changes sign across c

b. $f'(c) = 0$

c. $f''(c) = 0$ only

d. $f(c) = 0$

42 [Challenging] • Target: Second Derivative Test for Local Extrema • Cross-Lesson: 2.01 The Definition of the Derivative

When is the Second Derivative Test inconclusive at a critical point c?

a. When $f''(c) = 0$ or $f''(c)$ does not exist

b. When $f''(c) > 0$

c. When $f'(c) = 0$

d. When $f(c) = 0$

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43 [Challenging] • Target: Applied Optimization: Geometric Problems • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

A farmer has 200 meters of fencing to enclose a rectangular field adjacent to a straight river (no fence needed along the river). What dimensions maximize the area?

a. Length 100 m, Width 50 m

b. Length 80 m, Width 60 m

c. Length 120 m, Width 40 m

d. Square of side 66.7 m

44 [Challenging] • Target: Applied Optimization: Geometric Problems • Cross-Lesson: 3.01 The Chain Rule

What is the maximum volume in the box problem above?

a. 128 cm^3

b. 144 cm^3

c. 108 cm^3

d. 96 cm^3

45 [Challenging] • Target: Applied Optimization: Cost & Distance • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

A company determines that total cost for producing x units is $C(x) = 0.01x^2 + 4x + 100$. What production level minimizes average cost $\bar{C}(x) = \frac{C(x)}{x}$?

a. $x = 100$

b. $x = 50$

c. $x = 200$

d. $x = 400$

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